

Additionally, the employer can specify the experience level of workers they are looking for, choosing from three options: “I am willing to hire an inexperienced freelancer cheaply” (novice), “I prefer an intermediate level freelancer” (intermediate), or “I am willing to pay more for an experienced freelancer” (veteran). This information is presented to workers when they apply for jobs. However, it is important to note that the specified worker level is merely a statement of employer expectations. Novice workers can still apply for and potentially be hired for jobs aimed at higher levels.

The job posting metadata also includes other potential control variables, such as expected hours per week, and the desired worker's home country. These variables were not useful as control variables due to a large share of missing values.

Finally, when creating a job posting, the employers can also declare their expected project budget. We use the employer declared budget as a proxy measure for employers' willingness to pay for labor.

2.3. From job postings to skill-clusters with BERTopic modeling

We use BERTopic to form and name clusters of job postings with similar skill requirements. This process is done in five steps, which we explain and justify below.

Embeddings. We give a text description to each job posting by appending the pre-text ‘Expert in’ to its list of skills, which are lower-cased and unhyphenated. For instance, a job posting with skills ‘ghostwriting’, ‘writing’, ‘ebook’, ‘creative writing’, and ‘english’ becomes ‘Expert in ghostwriting, writing, ebook, creative writing, english’. We then embed these text descriptions into a 384-dimensional space using the sentence transformer “all-MiniLM-L6-v2”, which is recommended for short texts.

We prefer this approach over skill co-occurrence since skills are not constant over time. For example, while ‘news writing style’ and ‘news writing’ refer to the same or very similar skill, ‘news writing style’ appears as a skill between 2021–2022 but not in 2023, while ‘news writing’ is a skill present in 2022–2023, but not before. Network co-occurrence algorithms, which are commonly used to model skill networks (Anderson, 2017; Lukac, 2021; Demirci et al., 2023; Stephany and Teutloff, 2024), may struggle to see these two skills as overlapping. Transformer models like BERTopic are able to capture the overlap due to semantic similarity. In addition, compared to bag-of-words embedding approaches such as GloVe or Word2Vec (Pennington et al., 2014; Mikolov et al., 2013), transformers have the advantage of not being restricted to a predefined dictionary of words.

Dimensionality reduction The next step is to reduce the dimensionality of the embeddings using Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) (McInnes et al., 2018). This step is crucial because, in high-dimensional spaces, data points tend to become equidistant from each other, making it difficult for clustering algorithms to identify meaningful groupings (Aggarwal et al., 2001). Moreover, high-dimensional representations often contain noise or irrelevant features, which dimensionality reduction helps filter out. By focusing on the most important features, dimensionality reduction techniques enhance the quality of the clusters (Van Der Maaten et al., 2009).

Clustering We then cluster projects using Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). We adjust the minimum cluster size for HDBSCAN to 1000, as the BERTopic documentation recommends trying different values and setting parameters larger than 500 for datasets exceeding one million entries (Grootendorst, 2024b). We further use an outlier reduction function to decrease the number of uncategorized topics (Grootendorst, 2024a; McInnes and Healy, 2017).

We use HDBSCAN instead of K-means because K-means assumes clusters to be spherical and of roughly equal size. HDBSCAN, however, utilizes density within the embeddings, accommodating clusters of varying shapes (Bhattacharjee and Mitra, 2021). This approach is generally preferred for complex data distributions, such as text embeddings, and is recommended by the BERTopic documentation (Grootendorst, 2022). HDBSCAN also provides a hierarchy of clusters, allowing us to check cluster coherence manually.

This step resulted in 286 clusters. We manually checked the common skills and several job descriptions in each cluster and concluded that, at least intuitively, the clustering seems to produce meaningful results. Since BERTopic is stochastic in nature, the results depend on the initial, random allocation of topics to clusters. To test the robustness of our clusters, we ran the algorithm multiple times and also tried K-means clustering instead of HDBSCAN. The robustness checks are detailed in Appendix E.

Vectorization The clustering procedures end in the previous step. However, BERTopic facilitates giving intuitive names to each cluster with two more steps. These two steps do not affect the categorization of clusters into substitutable, complementary, or unaffected (see next section) and consequently do not affect our difference-in-difference estimation. Nonetheless, the names help interpret our results. The vectorization step converts text in clusters into numerical vectors, which creates a document-term matrix where each entry represents the frequency of a token in a document.

cTF-IDF Finally, to label each cluster, the most representative words are chosen via continuous term frequency-inverse document frequency (cTF-IDF). This process balances out frequent terms and removes those that are too frequent, ensuring that the most informative terms for each cluster are highlighted. This helps in assigning meaningful labels to the skill clusters.

In Appendix A, we provide a comparison of our clusters to the categories used by the online freelance platforms and additional arguments for our choice of clustering methods and robustness checks.

2.4. From skill clusters to AI exposure labels with GPT-4o

We apply the definitions from Eloundou et al. (2023) and Qiao et al. (2023) to group skill clusters into the following three categories of AI exposure:

Substitutable: A non-expert person can use Large Language Models to complete these tasks with a similar level of quality without the help of an expert.

Complementary: A non-expert cannot use Large Language Models to complete these tasks without significantly compromising the quality of execution. However, an expert freelancer can use Large Language models to reduce the time it takes them to complete these tasks by at least half and without significantly compromising quality.

Unaffected: While non-experts and experts may be able to use Large Language Models to help in these tasks, Large Language Models can only provide limited help. In other words, having access to a large language model will not reduce the time it takes an expert freelancer to complete the job by more than half.

These represent *ex-ante* definitions of the substitution, complemented and unaffected categories. While we conduct the manual and GPT-based labeling in 2024, we only consider LLM capabilities that were available by the time ChatGPT was released in November 2022 (for more details and showcase examples, see appendices B and C).

Manual labeling. To reliably classify skills clusters into these three groups, we combine several steps of manual analysis and GPT prompting as outlined in Fig. 1 sections two and three. First, two authors independently label all skill clusters. Reviewing and discussing the labels allows us to create a high-confidence set of 55 skill clusters for which we are confident about the AI exposure labels. This set includes 17 clusters in substitution, 14 in complementarity, and 24 in unaffected. However, since many of the skill clusters consist of highly specialized skills, reliable manual labeling would require expert knowledge in a variety of different occupational fields. Therefore, we rely on GPT-4o to extend our labels to the entire set of skill clusters using the high confidence set to optimize GPT prompts and benchmark the results.

Prompt engineering. To develop a prompt for GPT-4o that produces high-quality AI exposure labels for all skill clusters, we employ several validated prompt engineering best practices with the high confidence set serving as our *test set* on which to evaluate GPT output. As demonstrated by Brown et al. (2020), LLMs have been found to possess 'few-shot properties', i.e. their performance across most tasks increases when the prompt includes several demonstrative examples. Additionally, so-called 'chain-of-thought' prompting, whereby the user also gives examples of the logical steps that should lead to the correct answer, has been shown to significantly improve the complex reasoning capabilities of LLMs (Wei et al., 2022). We consider these insights by including an example for each category and a precise description of the reasoning behind the classification to the prompt. Further, as suggested by OpenAI (2023), we format our prompt such that the instructions are at the beginning and are separated through quotation marks from the context part. We further explicitly articulate our desired output format (JSON).

For each cluster, our input in the prompt includes the top ten skills as identified by cTF-IDF from BERTopic and the cluster's most frequent skills. We incorporate don't solely rely on the skills identified by cTF-IDF since those may not include relevant skills that are common across many clusters. The final best-performing prompt achieves 93 percent accuracy on the high confidence set. The full prompt text and examples for GPT output can be found in Appendix B. Using this final prompt to generate labels for all skill clusters results in 31 clusters assigned to substitution, 164 to complementarity, and 91 to unaffected.

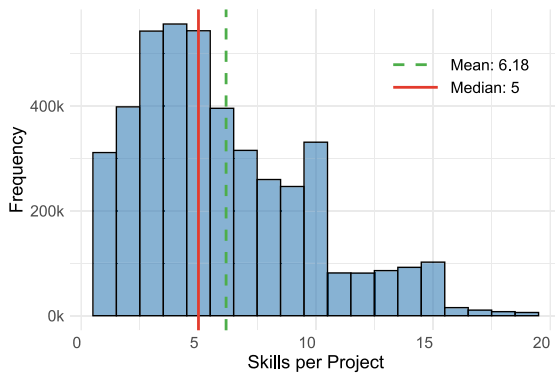
Consistency checks. To ensure consistent, high-quality labels, we manually review all GPT labels and the corresponding reasoning. Moreover, we use hierarchical clustering to aggregate the original 286 skill clusters into higher-level groups. Label coherence within the higher-level skill clusters corroborates the validity of the GPT labels. We change labels where the GPT reasoning is unconvincing and labels are not aligned within higher-level hierarchical skill clusters. Lastly, we manually merge small skill clusters following the hierarchical structure. While high granularity is beneficial when assigning AI exposure labels, small skill clusters are, at times, not stable over time. The online freelancing platform seems to periodically introduce new names for skills while discontinuing old ones. As a result, a few skill clusters suddenly disappear while other very similar skill clusters see sudden surges in their number of jobs. To smooth this noise, we merge *problematic* skill clusters into clusters with similar skills. This approach allows us to maximize the coherence of AI exposure labels while minimizing noise in the job count of skill clusters caused by the renaming of skills by the platform. We obtain a final set of 116 clusters, of which 12 are labeled as substitution, 59 as complementary and 45 as unaffected. Appendix section B provides additional information on how we create the AI exposure labels including details on manual labeling, prompts, label revision, and robustness.

2.5. Difference-in-differences and causal inference

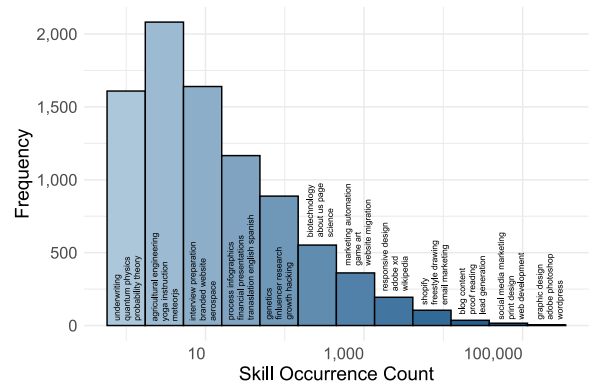
To estimate the causal effect of the ChatGPT launch on labor demand, we use a standard difference-in-differences design. Our control group consists of skill clusters where we assume ChatGPT cannot significantly aid in accomplishing tasks associated with jobs in those clusters. The treated clusters are labeled as either substitutable or complementary (examples of substitutable and complementary clusters can be found in Appendix C). We compare the two treated groups to the control group and estimate how the difference between them evolved since the launch of ChatGPT. More concretely, we express the log count of new job postings in cluster i in week t as:

$$\log(Postings_{it}) = \beta_0 + \beta_1 \text{Complementary}_i + \beta_2 \text{Substitutable}_i + \gamma \text{After}_t + \delta_1 (\text{Complementary}_i \times \text{After}_t) + \delta_2 (\text{Substitutable}_i \times \text{After}_t) + \epsilon_{it}. \quad (1)$$

After_t is a dummy variable that gets value 1 after the public launch of ChatGPT on the week of the 30th of November 2022 (and zero before that week). The binary variable Complementary_i gets value 1 if cluster i is labeled as Complementary, and analogously, the binary variable Substitutable_i has value 1 if the cluster is considered containing mainly skills relevant for tasks where humans can be substituted by LLMs. As sensitivity checks, we also estimate variations of Eq. (1) with cluster fixed and week fixed effects to account for seasonal variation in market labor demand and possible gaps in data collection.



(a) Distribution of the number of skills per project



(b) Distribution of skills

Fig. 2. (a) Distribution of the number of skill tags per job posting. (b) Overall distribution of the skill tags in the data set.

To estimate the effect at the skill cluster level we use the following regression model

$$\log(\text{Postings}_{it}) = \beta_0 + \beta_1 \text{Treated}_i + \gamma \text{After}_t + \delta_i (\text{Treated}_i \times \text{After}_t) + \epsilon_{it}, \quad (2)$$

where $\text{Treated}_i = 1$ whenever the cluster is in the substitutable or complementary categories and δ_i corresponds to the effect on skill cluster i .

The log transformation on the outcome variables of Eqs. (1) and (2) allows us to interpret the estimated δ 's as the percentage changes in postings caused by the launch of ChatGPT. To account for possible correlation within skill clusters, we cluster our standard errors at the skill-cluster level.

For the difference-in-differences model to capture the causal effect of the ChatGPT launch on labor demand, we need to make the standard parallel trends assumption, i.e. that in the absence of treatment, the difference between the two treatment groups and the control group would have remained unchanged. In our setting, this implies that we need to assume that the ChatGPT launch would not have had an effect on the demand for labor in the unaffected cluster. While this is ultimately untestable, we demonstrate that the control group and the two treatment groups evolved almost in parallel before the launch of ChatGPT.

There are some additional threats to our identification strategy that we cannot address. The platform in question may have made marketing decisions or pricing changes after the launch of ChatGPT. If these activities were concentrated on skill clusters labeled as substitutable or complementary, the difference-in-differences design might partly capture these non-ChatGPT effects. However, web analytics data from , indicates that monthly traffic on the platform has remained approximately constant around the ChatGPT launch date. Moreover, there were wide concerns about a labor shortage — i.e. a large number of unfilled vacancies — in the U.S. in months around the launch of ChatGPT. This could have led to an increased demand for platform labor. Given that all three groups showed similar trends before the ChatGPT launch, it is unclear why this business cycle effect would be stronger for skill clusters in the unaffected and complementary categories, though.

3. Results

3.1. Descriptive results

3.1.1. Skill requirements of job postings

Before examining the impact of LLMs on online labor markets, we present a descriptive analysis of our dataset. As illustrated in Fig. 2(a), most job postings list between three to eight skills, with a median of five skills per job posting. The ubiquity of skills across job postings is highly heterogeneous, as depicted in Fig. 2(b). Popular skills such as JavaScript, graphic design, and content writing appear in nearly a million job postings, highlighting their fundamental role in the gig economy. Conversely, specialized skills like English-Dutch translation, PowerPoint expertise, and deep neural network development are featured in approximately a thousand job postings, reflecting their niche demand. Additionally, highly specialized skills such as Montessori education or specific sign languages are mentioned only a handful of times, indicating a very limited but specific demand.

3.1.2. Embedding space and AI exposure labels

Our clustering and labeling methodology results in 116 skill clusters, of which 12 clusters are labeled as 'substitutable' (i.e., those types of jobs can be conducted by a non-expert with the help of LLMs), 59 as 'complementary' (i.e., those jobs in which the use of LLMs increases productivity substantially, but which still require expert knowledge) and 45 as 'unaffected' (i.e., those job types where LLMs do not help freelancers or where they lead only to a non-substantial productivity increase). Appendix C showcases examples of skill clusters, their representative skills, their assigned category, and the reasoning provided by GPT for the category assignment.

Table 1

Descriptive Statistics of the data per AI exposure category - Panel A: number of clusters, means, median and standard deviations. Panel B: distribution of expected project duration by category. Panel C - expected worker experience levels within categories.

Panel A: Descriptive Statistics of variable Job postings (logarithmic values in parentheses)				
Category	Number of clusters	Mean	Median	Standard deviation
Unaffected	45	451 (5.3)	208 (5.3)	707 (1.3)
Complementary	59	295 (5.2)	185 (5.2)	316 (1.1)
Substitutable	12	402 (5.2)	166 (5.1)	660 (1.2)
Panel B: Expected Duration Shares				
	Up to 3 weeks	3-9 weeks	9-18 weeks	18-52 weeks
Unaffected	48%	22%	10%	20%
Complementary	47%	25%	10%	18%
Substitutable	51%	22%	11%	16%
Panel C: Worker Experience Level Shares				
	Novice	Intermediate	Veteran	
Unaffected	9%	61%	30%	
Complementary	8%	58%	34%	
Substitutable	12%	60%	28%	

The substitutable category contains the fewest assigned clusters, while the complementary category has the most. This distribution is expected, as complementary skill clusters require only a subset of tasks to be influenced by LLMs, whereas substitutable clusters necessitate that most tasks be automatable. Consequently, the combinatorial possibilities for complementary skill clusters are larger, resulting in both more clusters and greater heterogeneity within this category (Arthur, 2010). This pattern is consistent with findings from Eloundou et al. (2023), who show that only 1.8% of tasks in the U.S. labor market can be fully automated, 18.5% of workers have more than 50% of their tasks exposed, but for about 70% of tasks only certain components could be completed by an LLM.

Fig. 3(a) illustrates how the transformer model embeds job postings based on their skill requirements and clusters those with similar skills together. These figures are a two-dimensional projection of the high-dimensional word embedding space. The first panel shows the eight largest clusters (i.e. clusters with the most job postings) within the substitutable category. These clusters are mostly related to writing and translation. Writing-related clusters such as proofreading (red), blog writing (light green), and script writing (purple) are close together on the right side of the embedding space. In contrast, translation topics such as German and Japanese translation are close together on the left side. The second and third panels show the eight largest clusters in the complementary and unaffected categories. These figures intuitively show that more similar clusters are placed closer together (e.g., in the complementary category, programming-related skills clusters are located close to each other).

Fig. 3(b) shows the most frequent skills required by job postings in each category. To a large extent, those match our expectations: In the complementary category, we frequently find skills related to software development (e.g. web development, web design, API) as well as references to specific coding languages or tools (e.g. javascript, html, css, php). This aligns with existing research (Moussiades et al., 2024) showing that ChatGPT can efficiently be used to significantly speed up coding. Within the substitutable category, we see different types of writing skills (e.g., blog writing, copywriting, article writing) as well as translation skills (e.g., French, English, Spanish to English translation). Lastly, within the unaffected category, we find many skills that are related to visual capabilities (e.g., graphic design, Adobe Photoshop, illustration), which could not be done with the release version of ChatGPT from November 30th 2022.

3.1.3. Descriptive statistics

After clustering the job postings and labeling the clusters, we aggregate the data on a weekly level. This data forms our analysis sample (Table 1). The average unaffected cluster has 451 job postings per week, the average complementary cluster has 295 job postings per week, and the average substitutable cluster has 402 new job postings per week. Across the three categories, means are much larger than medians, indicating a right-skewed distribution of job postings. After a log transformation, the means and medians of the variables are relatively close to one another.

In Panel B, we show the distribution of expected contract lengths across categories. These distributions are very similar to each other. The same applies to the desired worker experience level. In roughly 60% of the job postings, the employer has specified a preference for hiring a worker with intermediate experience. About 30% of job postings seek a veteran, and approximately 10% would consider hiring an inexperienced worker.

The overall number of job postings in the online labor market has grown substantially, as shown in Fig. 3(c). This time series shows the demand dynamics for popular skills, which are either substitutable (red), complementary to (green), or unaffected (blue) by LLMs. The demand for exemplary skills in the complementary and unaffected groups has grown substantially, while it showed a downward trend for skills in the substitutable category.